Integrative project

Zenith: Gamified Interactive Learning Platform for the Practice and Evaluation of Technical Skills

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General Project Objective

To design and develop an interactive web platform that complements Learning Management Systems (LMS), enhancing the educational process through the integration of real-time practice, gamification elements (streaks, badges, and scoring), and coder progress tracking. The goal is to increase motivation, participation, and knowledge retention by providing an innovative solution adaptable to any educational or corporate learning environment.

Specific Objectives

1. To design the architecture of a web platform that enables integration with Learning Management Systems (LMS).
2. To develop an interactive system that allows the execution of practical exercises within the platform, facilitating immediate learning through hands-on practice.
3. To implement a gamification system that includes usage streaks, badges, points, and levels in order to encourage continuous user participation.
4. To create modules of activities and interactive games that reinforce learning through playful dynamics and practical challenges.
5. To develop a user progress tracking system that allows visualization of statistics, achievements, and learning evolution.
6. To integrate a database capable of storing user information, progress data, scores, and achievements obtained during the use of the platform.
7. To design an intuitive and attractive interface that improves user experience and facilitates interaction with educational content.

Identified Problem

Traditional educational platforms often present several limitations in the learning process:

• They mainly provide static theoretical content.
• They offer limited opportunities for immediate practical interaction.
• They do not generate enough motivation for consistent platform usage.

• They lack engaging gamification elements.
• They do not effectively encourage consistency or active participation.

As a result, these limitations lead to:

• Low knowledge retention.
• Decreased motivation among learners.
• Course abandonment.
• Underutilization of the educational platform.

Proposed Solution

To develop a complementary platform that:

• Allows users to execute code within interactive windows.
• Integrates educational games related to the topics studied.
• Generates streaks based on continuous platform usage.
• Awards badges for achievements and milestones.
• Displays progress statistics.
• Allows users to visualize scores and levels (similar to Duolingo-style systems).
• Connects to databases (PostgreSQL) to store user progress.
• Is adaptable to any programming language or learning area.

Benefits

For Students

• Increased motivation.
• Immediate hands-on learning.
• Instant feedback.
• Sense of achievement.
• Clear progress tracking.

For Educational Institutions

• Higher student retention rates.
• Increased engagement and participation.
• Improved learning performance metrics.
• A more attractive and competitive learning platform.

For Companies

• Gamified internal training programs.
• Employee performance tracking.

• Interactive technical training.
• Adaptability to any area of knowledge.

Market Value

Currently, many Learning Management Systems such as:

- Moodle
- Blackboard
- Canvas
- Google Classroom

do not natively integrate:

• Real-time code execution.
• Deep video game–style gamification systems.
• Continuous reward and motivation systems.

This project has commercial value because it can:

• Be sold as a plugin for existing LMS platforms.
• Be implemented as a Software as a Service (SaaS) platform.
• Be offered to companies for internal employee training.
• Be customized for different industries (technology, languages, sales, etc.).

Differentiation from Other Platforms

What makes this proposal different:

1. Immediate practical learning directly within the platform.
2. Structured gamification (not only points, but also streaks, levels, and badges).
3. Adaptability to any learning domain.
4. A user experience similar to educational mobile game applications.
5. A focus on behavioral motivation based on learning psychology.

While other platforms simply teach,
this platform aims to make users want to return every day.

Adaptability to Any Learning Area

The platform structure allows implementation in:

- Programming courses
- Language learning
- Mathematics
- Sales training
- Corporate training programs
- Corporate onboarding processes
- Technical skills development programs